

① 3 2.

76-20

82	82	85	94	97	84	90	92	76	75
77	75	92	94	80	92	97	84	75	72
76	79	83	82	78	87	87	78	77	88
86	77	89	90	87	88	89	76	82	79
75	87	78	72	87	86	72	89	90	89
79	82	89	76						

$n = 52$

x_i	75	76	77	78	79	80	87	82	83
n_i	4	3	6	4	3	7	3	2	7

x_i	84	85	86	87	88	89	90	90	92
n_i	2	7	2	4	1	5	3	1	3

x_i	94
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n_i	2
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$$\begin{aligned}
 1. \quad \bar{x}_6 &= \frac{1}{n} \sum_{i=1}^k x_i n_i = \frac{1}{52} (75 \cdot 4 + 76 \cdot 3 + \\
 &+ 77 \cdot 6 + 78 \cdot 4 + 79 \cdot 3 + 80 + 81 \cdot 3 + 82 \cdot 2 + \\
 &+ 83 + 84 \cdot 2 + 85 + 86 \cdot 2 + 87 \cdot 4 + 88 + 89 \cdot 5 + \\
 &+ 90 \cdot 3 + 91 \cdot 2 + 92 \cdot 3 + 94 \cdot 2) = \frac{4337}{52} \approx \\
 &\approx 83,29
 \end{aligned}$$

$$2. \quad D_6 = \frac{1}{n} \sum_{i=1}^k (x_i - \bar{x}_6)^2 n_i = \frac{1}{n} \sum_{i=1}^k x_i^2 n_i - (\bar{x}_6)^2$$

$$\begin{aligned}
 \frac{1}{52} \sum x_i^2 n_i &= \frac{1}{52} (75^2 \cdot 4 + 76^2 \cdot 3 + 77^2 \cdot 6 + 78^2 \cdot 4 + \\
 &+ 79^2 \cdot 3 + 80^2 + 81^2 \cdot 3 + 82^2 \cdot 2 + 83^2 + 84^2 \cdot 2 + \\
 &+ 85^2 + 86^2 \cdot 2 + 87^2 \cdot 4 + 88^2 + 89^2 \cdot 5 + 90^2 \cdot 3 + \\
 &+ 91^2 \cdot 2 + 92^2 \cdot 3 + 94^2 \cdot 2) = \frac{362567}{52} \approx 6972,33
 \end{aligned}$$

$$D_6 = 6972,33 - (83,29)^2 \approx 35,77$$

$$3. \quad s^2 = \frac{n}{n-1} D_6 = \frac{52}{51} 35.12 \approx 35.8$$

$$\sigma_6 = \sqrt{s^2} = \sqrt{35.8} \approx 5.98$$

1. нормальний розподіл $\Rightarrow$

$$\bar{a} = \bar{x}_6 = 83.29$$

$$\bar{\sigma} = \sqrt{s^2} = 5.98$$

2. норм. розподіл — симетричний.

$$\bar{a} = \bar{x}_6 = 83.29$$

$$\bar{\sigma} = \sqrt{s^2} = 5.98$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-a)^2}{2\sigma^2}} =$$

$$= \frac{1}{5.98 \sqrt{2\pi}} e^{-\frac{(x-83.29)^2}{.71.52}}$$

